Ruchitha D

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SUMMARY

Data Analyst | Data Scientist | Python Developer with hands-on experience in building scalable data-driven applications, machine learning models, and backend systems. Strong expertise in data analysis, predictive modeling, statistical analysis, and API development. Proficient in Python, SQL, and BI tools with experience in deploying ML models and developing end-to-end data pipelines. Skilled in transforming raw data into actionable insights, improving business decision-making, and optimizing system performance.

TECHNICAL SKILLS

Data Analytics: Data Analysis, Exploratory Data Analysis (EDA), Statistical Analysis, Predictive Analytics, Data Modeling, Hypothesis Testing, Time Series Analysis, Data Mining, Data Wrangling, Feature Engineering, A/B Testing

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Neural Networks, CNN, Model Evaluation, Model Deployment, Hyperparameter Tuning, NLP, Computer Vision

Data Science and Analytics: Data Science, Machine Learning, Scalability, Quality control, Infrastructure, Credit risk

Programming Languages and Development: Python, C, SQL, JavaScript, OOP, Data Structures & Algorithms

Tools and Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, NLTK, Keras

Visualization and BI: Power BI, Tableau, Dashboard Development, Data Visualization, Excel Analytics

Backend and Deployment: Flask, FastAPI, REST APIs, Firebase

Databases: MySQL, MongoDB

Development Practices: Agile Methodology, Version Control, Git, GitHub

Cloud: AWS

EXPERIENCE

Data Analyst / AI Integration Intern — **Feb 2026 – Jun 2026 (Present)**
TechnoFly Solutions — *Bengaluru (On-site)*

- Working on real-time data analytics projects involving structured and semi-structured datasets to support operational and business decision workflows
- Designing and implementing data preprocessing pipelines including data cleaning, transformation, normalization, and feature engineering for machine learning models
- Performing exploratory data analysis (EDA) to identify trends, anomalies, and performance improvement opportunities across business datasets
- Developing predictive analytics models using supervised learning techniques to support forecasting and performance optimization tasks
- Building interactive dashboards and reporting systems using Power BI and Python visualization libraries to track KPIs and business metrics
- Writing optimized SQL queries for data extraction, aggregation, and reporting across relational databases
- Assisting in deployment of machine learning models into web applications using Flask/FastAPI for real-time inference workflows
- Conducting model evaluation and performance tuning using appropriate statistical and validation techniques

- Collaborating with software development teams to integrate analytical solutions into production-level systems
- Supporting API-based data pipelines for real-time analytics and automation use cases
- Documenting model workflows, analytical findings, and system improvements for internal technical and business stakeholders

Web Development Inten **Mar 2025 – May 2025**
Smart Learn EdTech

- Architected high-throughput REST APIs with FastAPI, enabling real-time asynchronous data processing and seamless frontend-backend integration
- Achieved 20% reduction in system latency by optimizing database queries, implementing caching layers, and improving API response efficiency
- Actively contributed to Agile SDLC processes including sprint execution, peer code reviews, and feature deployment within a collaborative 5-engineer team
- Implemented Docker-based containerization and cloud deployment workflows to support scalable, fault-tolerant application infrastructure

PROJECTS

Campus Utility Analytics Platform

- Enhanced a full-stack campus management system integrating real-time data monitoring modules
- Built analytics dashboards for transportation tracking and resource utilization
- Led a 4-member team; project awarded 1st place in departmental hackathon

Financial Analytics Expense Tracker

- Designed predictive budgeting analytics system to analyze user spending behavior
- Applied data visualization dashboards to improve financial decision insights
- Expanded secure backend services using FastAPI and MySQL

Real-Time Computer Vision Identity Verification

- Pioneered advanced image preprocessing pipeline using OpenCV and TensorFlow
- Improved object detection accuracy by 15% and reduced false positives by 22% in low-light conditions.

GPS-Based Transport Analytics System

- Improved real-time location analytics system using Google Maps API
- Created interactive visualization interface for operational tracking insights

E-Commerce Behavioral Data Module

- Enforced user activity tracking and analytics-driven recommendation logic
- Designed database queries to optimize transaction processing performance

Breast Cancer Prediction

- Developed a Convolutional Neural Network (CNN) model for breast cancer classification using medical imaging data, achieving high accuracy through image preprocessing, data augmentation, and hyperparameter tuning
- Implemented end-to-end pipeline including data preprocessing, model training, validation, and performance evaluation (accuracy, precision, recall, F1-score), improving early-stage cancer detection capability

EDUCATION

Navkis College of Engineering **VTU**
B.E in Artificial Intelligence And Data Science *2022–2026*

CERTIFICATIONS

- **NPTEL** – Python for Data Science (Distinction), Database Management System, Data Structures and Algorithm Design
- **Infosys Springboard** – Introduction to Data Science
- **EXCELR** – Full Stack Development
- **Workshops** – Power BI, Industrial AI, ChatGPT & AI Tools